

Hemanth kumar

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Summary:

Data Engineer with 6 years of experience building large-scale data pipelines and AI-driven solutions in the banking and financial sector. I specialize in **Apache Spark, PySpark, and Azure Databricks**, designing scalable data workflows using **Python, SQL, and Delta Lake**. My work combines **data engineering and AI automation**, integrating **Azure AI, OpenAI, and Power Platform tools** like **Power BI and Copilot Studio** to improve forecasting, reconciliation, and decision-making processes. I've also worked extensively with **Informatica, Oracle, and Azure Data Lake**, delivering compliant and high-performance systems aligned with **Basel III** and **BCBS 239** standards. I'm passionate about turning complex data into actionable insights that connect business goals with reliable, cloud-ready technology.

TECHNICAL SKILLS:

Category:	Technologies & Tools:
Programming Languages	Python, PySpark, SQL, Scala Linux/Unix
AI / ML Frameworks	Azure AI, OpenAI, PyTorch (sometimes)
Automation & Integration	Ms Power Platform, Power automate, Copilot Studio
Big Data Technologies	Apache Hadoop, Spark, Databricks, PySpark, Apache Kafka
Databases	MySQL, MongoDB, Cassandra, HBase
Data Warehousing	Amazon Redshift, Snowflake, Azure Synapse
ETL Tools	Databricks, Informatica, HDInsight, Azure Dataflow, Airflow
Data Modeling	Star Schema, Snowflake Schema
Cloud Technologies	AWS (EC2, S3, Lambda, EMR, Glue, Redshift), Azure (Data Factory, Data Lake, Databricks, SQL Database, Cosmos DB)
DevOps & CI/CD	GitHub, GitLab, Shell Scripting, Azure Logic Apps
Robotic Process Automation (RPA)	UIPath, Power Automate Desktop
Data Visualization	Tableau, Power BI
Containerization & Orchestration	Docker, Kubernetes
Data Security	Encryption, Data Masking, Compliance with GDPR/CCPA
Key Methodologies	Real-time Data Processing, Data Migration, Cloud-Native Solutions, Performance Optimization (Indexing, Query Optimization, Caching)

PROFESSIONAL EXPERIENCE:

Fifth Third Bank, Dallas, USA

Role: Data Engineer (Contract)

06/2024 – Till date

Technologies: Azure Cloud (ADLS, ADF, Databricks), Azure Devops, Azure AI, Open AI, SQL, Snowflake, Python, RPA, PyTorch, Alation, Pyspark, Apache Kafka, Oracle, DataStage

- Developed scalable **ETL workflows using Informatica** to extract and cleanse financial data from **Oracle** databases, integrating results into **Azure Data Lake** for analytics and risk modeling.
- Designed and implemented **AI-powered anomaly detection and forecasting models** using **Azure Machine Learning, Python, and Azure OpenAI**, improving accuracy in financial data reconciliation and operational reporting.
- Developed **Copilot-integrated Power Automate workflows** to streamline reconciliation and exception-handling tasks, reducing manual data validation efforts by **over 25%**.
- High reliability and adherence to internal SLAs were ensured by designing end-to-end data intake pipelines in Azure Databricks to process daily loan, credit, and customer data from core banking systems.
- Implemented **incremental load strategies and Delta Lake merges** to capture only new or changed records, minimizing data duplication and improving processing efficiency by 35%.
- Designed **data validation and reconciliation frameworks** to ensure accuracy in sensitive financial datasets used for credit risk scoring, capital adequacy, and regulatory submissions.
- Built **end-to-end data integrations between Azure Databricks, Power BI, and Dynamics 365 Finance** to surface machine learning predictions and automate KPI monitoring for risk and forecasting workflows.
- Partnered with business analysts and compliance teams to translate **banking domain rules** (like Basel III and BCBS 239) into automated data quality and transformation logic.
- Collaborated with cross-functional teams to implement **Python- and PySpark-based transformations**, optimizing performance across diverse data sources and large-scale datasets.
- Integrated **Azure DevOps CI/CD pipelines** for Databricks notebooks, streamlining code promotion between development and production environments while ensuring version control.
- Managed **metadata, data lineage, and auditing** to improve transparency and meet data governance standards required by banking regulators.
- Built **parameterized SQL and Databricks jobs** for recurring reporting tasks, reducing manual intervention and improving reporting turnaround times.
- Ensured all AI and automation solutions adhered to **enterprise data governance, compliance, and security standards**, maintaining transparency and traceability across operational workflows.
- **Automated data forecasting pipelines in Azure Databricks** by integrating time-series models with financial transaction data, providing early visibility into loan performance and liquidity trends for decision-makers.
- By using Microsoft Power Platform and Copilot Studio teams to prototype AI-driven workflow assistants that guided analysts through data quality checks and reconciliation steps, improving process accuracy and user productivity.
- Participated in **Agile sprints using JIRA**, contributing to design discussions, task estimation, and sprint reviews.

USAA, Dallas, USA

Role: Data Engineer (Contract)

02/2023–05/2024

Technologies: Python, PySpark, AWS EMR, AWS Glue, AWS Lambda, Amazon EKS, Apache Kafka, Amazon Kinesis, Amazon Redshift, Jenkins, GitLab, AWS S3, Snowflake

- Worked on developing, debugging, testing, deploying and maintaining the data pipelines along with third-party jars, packages in all environments.
- Develop high-level and detailed solution architectures for data engineering projects, ensuring alignment with business objectives and scalability.
- Design solutions using big data technologies like PySpark over **Amazon EMR, AWS EKS, AWS Glue**, Pandas dataframe over **AWS Lambda, DynamoDB**, DMS and others. Real-time data processing using services like **Amazon Kinesis**, Apache Kafka, Apache Flink.
- Design and optimize data warehouses using services like Amazon Redshift and implement strategies to optimize query performance for large datasets
- Design and implement data models that meet business needs and ensure efficient data storage and retrieval.
- Set up **monitoring and logging** for data pipelines and warehouses. Diagnose and resolve issues in data pipelines and architectures
- Effectively communicate technical decisions, challenges, and solutions to both technical and non-technical stakeholders.
- Led the design and implementation of complex **ETL processes** using SQL Server Integration Services (SSIS), optimizing data flow and transformation logic to enhance performance and reliability.
- Extensive hands-on experience leveraging PySpark, Spark with Scala and Java on **Amazon EMR, Hadoop** with Java for large-scale data processing, ensuring efficient ETL workflows and optimal resource utilization. Data ingestion using Apache Sqoop, Python **JDBC application, Spark connectors**.
- Proficient in designing and implementing data pipelines on AWS, utilizing services such as **Amazon S3** for scalable and cost-effective storage, AWS Step Functions for orchestrating workflows, and **Amazon Redshift** for high-performance data warehousing, Databricks, **Elasticsearch**, Kibana
- Developed and deployed data processing pipelines using Python, **AWS Lambda**, and Snowflake, and built interactive data visualization dashboards
- Optimized enterprise data systems by ensuring seamless database operations, improving fault tolerance, and enhancing data processing efficiency, supporting real-time and batch data workflows across various platforms.

HCL, Hyderabad, India

Role: Data Engineer (Contract)

08/2018 –12/2021

Technologies: Python, PySpark, Apache Spark, Hive, HiveQL, Kafka, Kinesis, AWS (S3, EMR, Glue, Lambda, Redshift), Azure Data Lake, Azure SQL, Jenkins, GitLab CI, Kubernetes, Tableau, Power BI, HDFS

- Built and maintained high-volume batch data pipelines using **Hive, Apache Spark**, and **HDFS**, improving job efficiency and enabling near real-time analytics support.
- Wrote complex **HiveQL** queries to process multi-terabyte datasets, reducing query execution times and enhancing overall data pipeline throughput by 30%.
- Integrated **Hive** with Spark to leverage distributed processing, significantly reducing data preparation time for reporting and modelling teams.
- Tuned Python and Scala-based Spark jobs, improving runtime performance by 40% and minimizing compute cost on **AWS EMR** clusters.

- Designed and deployed real-time data streaming pipelines using **Apache Kafka** and **Amazon Kinesis**, enabling near-instant data ingestion and application monitoring.
- Improved data workflow speed by optimizing **Apache Oozie** jobs and restructuring Hive scripts, resulting in a 50% latency reduction in Java-integrated systems.
- Managed secure, scalable data storage using **Amazon S3**, **Redshift**, and **Azure Data Lake**, with access controls and encryption for compliance requirements.
- Set up and managed containerized data services with **Kubernetes**, supporting workload scaling and minimizing service downtime.
- Automated deployment and testing processes using **Jenkins** and **GitLab CI**, streamlining the release cycle across development, staging, and production environments.
- Developed modular Python scripts for data transformation, validation, and cleansing, enabling faster development and reusable logic across ETL workflows.
- Optimized **Redshift** queries and schema designs to accelerate report generation and reduce dashboard load times.
- Managed **Azure SQL** environments for secure, reliable access to reporting data across business units in different time zones.
- Delivered dashboards using **Power BI**, allowing business teams to access near real-time data insights.
- Monitored ETL jobs using **AWS CloudWatch** and configured alerting with **SNS**, ensuring prompt resolution of job failures or data issues.
- Collaborated with cross-functional teams to align data architecture with analytics goals, supporting both batch and streaming use cases in production.

EDUCATION: Master's in computer and information science — Southern Arkansas University, USA 2023

Bachelor of Engineering in Civil Engineering — Sathyabama University, India 2019